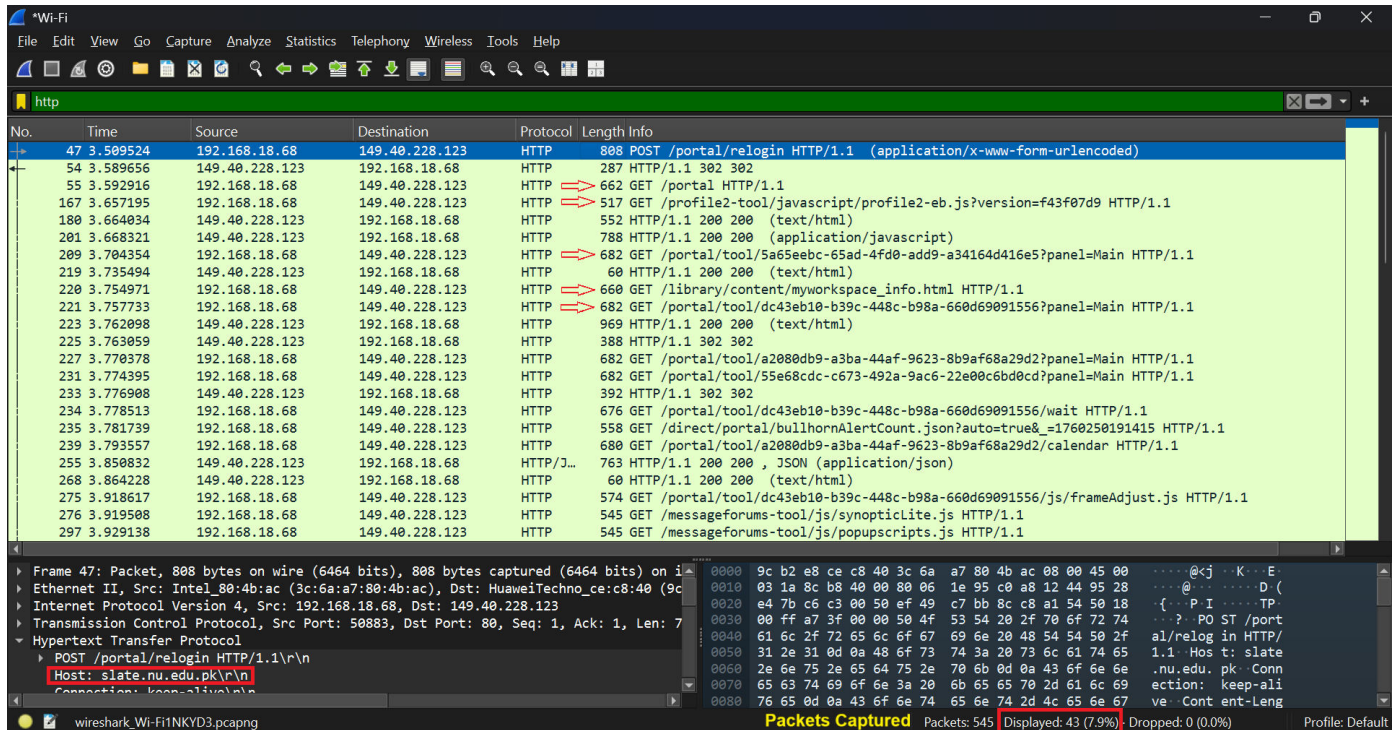


CNET Assignment 1

Question 3

1. Capture Live HTTP, FTP, and DNS Packets (at least 5 packets)

- Use Wireshark to filter and capture packets for:
 - HTTP GET requests



- DNS Query responses

The image shows a Wireshark capture of DNS traffic. The packet list pane displays several DNS queries and responses. A Windows PowerShell window is overlaid, showing the execution of a ping command to 8.8.8.8. The output shows successful pings with response times between 37ms and 56ms.

No.	Time	Source	Destination	Protocol	Length	Info
64	16.879156	192.168.18.68	192.168.18.1	DNS	75	Standard query 0xaf3d HTTPS ssl.gstatic.com
65	16.879905	192.168.18.68	192.168.18.1	DNS	75	Standard query 0xda9c AAAA ssl.gstatic.com
66	16.880470	192.168.18.68	192.168.18.1	DNS	75	Standard query 0x585d A ssl.gstatic.com
67	16.896924	192.168.18.1	192.168.18.68	DNS	135	Standard query response 0xaf3d HTTPS ssl.gstatic.com SOA ns1.google.com
68	16.896924	192.168.18.1	192.168.18.68	DNS	103	Standard query response 0xda9c AAAA ssl.gstatic.com AAAA 2a00:1450:4018:80c::2003
69	16.897480	192.168.18.1	192.168.18.68	DNS	91	Standard query response 0x585d A ssl.gstatic.com A 216.58.209.131
150	21.873212	fe80::6a19:539:2bd2...	fe80::1	DNS	114	Standard query 0x8170 A p2p-dxbl.discovery.steamserver.net
151	21.885466	fe80::1	fe80::6a19:539:2bd2...	DNS	146	Standard query response 0x8170 A p2p-dxbl.discovery.steamserver.net A 185.25.183.36 A 185.25.183.5

Windows PowerShell output:

```
PS C:\Users\ DELL > ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=38ms TTL=114
Reply from 8.8.8.8: bytes=32 time=40ms TTL=114
Reply from 8.8.8.8: bytes=32 time=56ms TTL=114
Reply from 8.8.8.8: bytes=32 time=37ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 37ms, Maximum = 56ms, Average = 42ms
```

FTP file transfer data

The image shows a Wireshark capture of FTP traffic. The packet list pane displays a series of FTP commands and responses, including login attempts, directory listings, and file uploads. A packet detail pane shows the structure of an FTP response packet, including the response code, message, and arguments.

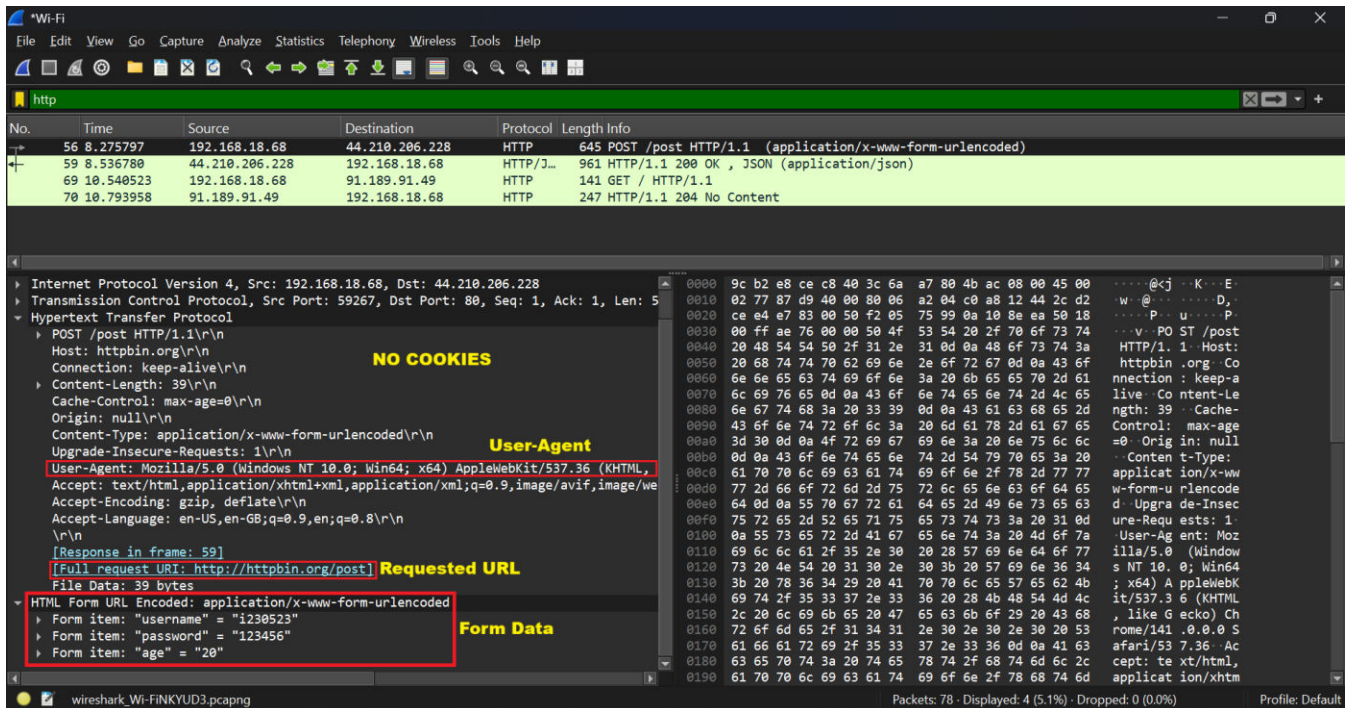
Time	Source	Destination	Protocol	Length	Info	
112046	647.195465	35.209.95.242	192.168.18.68	FTP	98	Response: 331 User dlpuser OK. Password required
112437	688.656920	192.168.18.68	35.209.95.242	FTP	86	Request: PASS rNrKYTX9g7z3RgJRMxwUGHbeu
112455	694.335703	35.209.95.242	192.168.18.68	FTP	91	Response: 530 Login authentication failed
112612	707.373909	192.168.18.68	35.209.95.242	FTP	60	Request: QUIT
112618	707.659078	35.209.95.242	192.168.18.68	FTP	125	Response: 221-Goodbye. You uploaded 0 and downloaded 0 kbytes.
112831	728.505541	194.108.117.16	192.168.18.68	FTP	229	Response: 220-Welcome to test.rebex.net!
112832	728.551987	192.168.18.68	194.108.117.16	FTP	68	Request: OPTS UTF8 ON
112834	728.835564	194.108.117.16	192.168.18.68	FTP	87	Response: 200 Enabled UTF-8 encoding.
112840	732.206596	192.168.18.68	194.108.117.16	FTP	65	Request: USER demo
112841	732.418390	194.108.117.16	192.168.18.68	FTP	134	Response: 331 Anonymous login OK, send your complete email address as your password.
112852	735.334730	192.168.18.68	194.108.117.16	FTP	69	Request: PASS password
112860	735.498014	194.108.117.16	192.168.18.68	FTP	86	Response: 230 User 'demo' logged in.
114182	841.291205	192.168.18.68	194.108.117.16	FTP	82	Request: PORT 192,168,18,68,219,158
114183	841.476686	194.108.117.16	192.168.18.68	FTP	103	Response: 425 Connections to other hosts not allowed.
114326	856.416184	192.168.18.68	194.108.117.16	FTP	82	Request: PORT 192,168,18,68,219,159
114327	856.630641	194.108.117.16	192.168.18.68	FTP	103	Response: 425 Connections to other hosts not allowed.
114705	896.928538	192.168.18.68	194.108.117.16	FTP	81	Request: PORT 192,168,18,68,210,98
114706	897.187784	194.108.117.16	192.168.18.68	FTP	103	Response: 425 Connections to other hosts not allowed.
114707	897.225691	192.168.18.68	194.108.117.16	FTP	81	Request: PORT 192,168,18,68,210,99
114708	897.488436	194.108.117.16	192.168.18.68	FTP	103	Response: 425 Connections to other hosts not allowed.
114709	897.523435	192.168.18.68	194.108.117.16	FTP	60	Request: QUIT
114710	897.795098	194.108.117.16	192.168.18.68	FTP	80	Response: 221 Closing session.

Packet detail for Transmission Control Protocol (FTP):

```
220-Welcome to test.rebex.net!\r\n
Response code: Service ready for new user (220)
Response arg: Welcome to test.rebex.net!
See https://test.rebex.net/ for more information and terms of use.\r\n
220 If you don't have an account, log in as 'anonymous' or 'ftp'.\r\n
[Current working directory: ]
```

2. Analyze a captured HTTP request

- Identify the requested URL, User-Agent, and cookies.



○ Explain how HTTPS protects sensitive data compared to HTTP.

Ans: HTTPS uses TLS/SSL encryption to secure communication between client and server.

The HTTPS:

- Encrypts all the form data such as username, password, etc.
- Authenticates server's identity through digital certificates.
- Prevents attackers from modifying information during data transmission.

Q 4

(i)

Transmission Rate (R) = 350 Mb/s

Object size (L) = 250,000 bits

Request Rate (λ) = 35 req/s

Avg internet-side round trip (T_{internet}) = 2.5 sec

(i) Total avg response time (T_{total}) = ?

$$\begin{aligned}\text{Data rate (Arrival rate)} &= 250000 \times 35 \\ &= 8.75 \times 10^6 \text{ bits/s} \\ &= 8.75 \text{ Mb/s}\end{aligned}$$

$$\text{Utilization } (\rho) = \frac{\text{Arrival rate}}{\text{Transmission rate}}$$

$$= \frac{8.75 \text{ Mb/s}}{350 \text{ Mb/s}}$$

$$= 0.025$$

$$= 2.5\%$$

$$\begin{aligned}\text{Access delay } (T_{\text{access}}) &= \frac{L/R}{1-\rho} \\ &= \frac{250000/(350 \times 10^6)}{1-0.025}\end{aligned}$$

$$= 0.73 \times 10^{-3}$$

$$\therefore T_{\text{total}} = T_{\text{access}} + T_{\text{internet}}$$

$$= (0.73 \times 10^{-3}) + 2.5$$

$$= 2.50073$$

$$\boxed{T_{\text{total}} \approx 2.5 \text{ Sec}} \text{ Ans.}$$

(ii)

$$\text{Miss rate} = 0.7$$

$$\text{Response time of cache} = 0.01$$

$$\Rightarrow \text{Hit rate} = 1 - 0.7 \\ = 0.3$$

$\Rightarrow$ Total response time (with cache):

$$T_{\text{total}(c)} = (\text{Miss rate} * T_{\text{total}}) + (\text{HitRate} * \text{Response cache time}) \\ = (0.7 * 2.50073) + (0.3 * 0.01)$$

$$T_{\text{total}(c)} = 1.753 \text{ sec} \text{ Ans.}$$

Q 5

Data:

$$\text{Buffer max capacity } (b_{\text{max}}) = 50\,000 \text{ packets}$$

$$\text{Packet size } (p_{\text{size}}) = 1000 \text{ bytes}$$

$$\text{Ongoing link} = 100 \text{ Mbps}$$

$$\text{Arrival link} = 150 \text{ Mbps}$$

$$t = 10 \text{ s}$$

1- Time for buffer overflow = ?

2- Packets dropped = ?

3- Advanced mechanisms (2) = ?

Sol:

$$\therefore 1 \text{ byte} = 8 \text{ bits}$$

$$b_{\text{max}} = 50\,000 \times (1000 \text{ bytes}) \times 8 \text{ bits} \\ = 400 \times 10^6 \text{ bits} \\ = 400 \text{ Mbits}$$

$$\text{Buffer overload} = (\text{Arrival link}) - (\text{Ongoing link})$$

$$= 150 - 100$$

$$V_{\text{overload}} = 50 \text{ Mbps}$$

$$\begin{aligned}\text{Overloaded data in } t \text{ secs} &= b_{\text{overload}} \times t \\ &= (50 \times 10) \text{ (Mbps} \times \text{s)} \\ &= 500 \text{ Mbits}\end{aligned}$$

$$\begin{aligned}\text{Time for buffer overflow} &= \frac{b_{\text{max}}}{V_{\text{overload}}} \\ &= \frac{400 \text{ Mbits}}{50 \text{ Mbps}} \\ 1. \quad \boxed{t_{\text{overflow}} = 8 \text{ s}} &\rightarrow \underline{\text{Ans.}}\end{aligned}$$

$$\begin{aligned}\text{Data dropped} &= (500 - 400) \text{ Mbits} \\ &= 100 \times 10^6 \text{ bits} \\ &= (100 \times 10^6) \div 8 \\ &= 12.5 \times 10^6 \text{ bytes}\end{aligned}$$

$$\text{Packets dropped} = (12.5 \times 10^6) \div 1000$$

$$2. \quad \boxed{= 12500 \text{ packets}} \underline{\text{Ans.}}$$

3- Advanced Mechanisms:

① Priority Queueing:

- Router maintains multiple queues.
- Different traffic goes into different queues.
- High priority packets get transmitted.
- When buffer fills, it drops low priority packets, first.

② Congestion Window Adjustment (At TCP Level):

- TCP sender maintains a congestion window (packets it can send)
- Starts small, then gradually increases.
- If packet loss detected, it immediately reduce window size.
- If no loss occurs, it gradually increases again.